

Please attach this cover sheet to your completed homework

Homework 6 (Ch. 12.5 – 12.6)

ECE 311 – Hang Gao

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Problem	Course outcome	Grade
12.17	1.c.	/20
12.18 (a)	2.c.	/10
12.18 (b)	1.c.	/10
12.18 (c)	1.c.	/10
12.18 (d)	1.c.	/10
12.19 (a)	2.c.	/10
12.19 (b)	1.c.	/10
12.22	1.c.	/20
TOTAL:		/100

ABET Assessed Outcomes

1.c. Use appropriate power system models to formulate solutions.

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2.c. Analyze power systems with safety, social, environmental, and economic factors and constraints.

Homework #6

Due date: Sunday, March 23rd by 11:59 PM (Electronically – Canvas)

- 12.17** A synchronous generator connected directly to an infinite bus is generating at its maximum real power. The line-to-line voltage of the infinite bus is 480 V and the synchronous reactance of the generator is $5\ \Omega$. Compute the reactive power of the generator.
- 12.18** A 1 GW synchronous generator is connected to an infinite bus through a transmission line. The synchronous reactance of the generator is $9\ \Omega$ and the inductive reactance of the transmission line is $3\ \Omega$. The infinite bus voltage is 110 kV. If the generator delivers its rated power at unity power factor to the infinite bus, compute the following:
- Terminal voltage of the generator
 - Equivalent field voltage
 - Real power output at the generator terminals
 - Reactive power output at the generator terminals
- 12.19** A synchronous generator is connected directly to an infinite bus. The voltage of the infinite bus is 15 kV. The excitation of the generator is adjusted until the equivalent field voltage E_f is 14 kV. The synchronous reactance of the machine is $5\ \Omega$. Compute the following:
- Pullover power (maximum power)
 - Equivalent excitation voltage that increases the pullover power by 20%
- 12.22** A synchronous motor excited by a 5 kV source has a synchronous reactance of $5\ \Omega$. The motor is running without any mechanical load (i.e., the output real power is zero). Ignore all losses and calculate the equivalent field voltage E_f that delivers 3 MVAR to the source. Also, draw the phasor diagram.